



National University
Polyclinics

PARKINSON'S DISEASE

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INTRODUCTION

Importance

Parkinson's Disease (PD) is a chronic progressive neuro-degenerative disorder and is a common cause of parkinsonism. PD tends to manifest initially with motor symptoms (related to a loss of dopaminergic neurons in the substantia nigra). In the later stages, non-motor features appear (related to neuronal loss in non-dopaminergic areas of brain):

- a) Autonomic dysfunction
- b) Falls
- c) Sleep disturbances
- d) Cognitive abnormalities

The pathogenetic mechanism of PD is still not fully understood. It is likely a combination of genes, environmental toxins, and free radicals. Pathological features include degeneration of dopaminergic neurons in the substantia nigra pars compacta coupled with intracytoplasmic inclusions known as Lewy bodies. Neurodegeneration and Lewy bodies can also be found in the locus ceruleus, nucleus basalis, hypothalamus, cerebral cortex, cranial nerve motor nuclei, and central and peripheral components of the autonomic nervous system.

It is believed that the neurodegenerative process in PD is a combination of both cell-autonomous and non-cell-autonomous mechanisms. Proposed cell-autonomous mechanisms include alterations in mitochondrial bioenergetics, dysregulation of calcium homeostasis, and impaired turnover of mitochondria. The proposed non-cell-autonomous mechanisms involve prion-like behaviour of misfolded proteins and neuroinflammation. This suggests that cell death in Parkinson's disease is caused by a multifactorial cascade of pathogenic events.

A significant number of patients diagnosed with PD show alternative diagnoses at autopsy, such as multiple systems atrophy, progressive supranuclear palsy and cerebrovascular disease.

Epidemiology

According to a community-based study in 2004, the prevalence of PD in Singapore was 0.3% of the population aged 50 years and above which represents a significant burden of disease to patient and family.

FEATURES AND DIAGNOSIS

PD remains a clinical diagnosis as there are currently no definitive diagnostic markers for PD and investigations are done to exclude / further assess other causes of parkinsonism. The clinical diagnosis of PD is made based on the presence of characteristic features, and the exclusion of alternative causes of parkinsonism.

The Movement Disorders Society 2015 Clinical Diagnostic Criteria for PD recommends the following approach:

- a) Confirm the presence of parkinsonism.
- b) Exclude atypical features that suggest alternative diagnoses.
- c) Supportive prospective criteria for PD.

Diagnosis

STEP ONE: Confirm Parkinsonism

Parkinsonism is defined by: A) bradykinesia; B) rest tremor and/or rigidity.

1. Bradykinesia. Mandatory feature in PD

- a) Slowed movements plus decrement in amplitude / speed or progressive hesitation / halts as movements are continued.
- b) Tested by getting patient to perform repetitive movements, e.g., finger tapping, alternating pronation & supination of forearm, opening & closing of fists.

2. Rest tremor

- a) Distal “pill-rolling”, 3-5 Hz. 50-70% patients with PD (no tremor does not mean no PD).
- b) Best detected with limb fully supported against gravity.
- c) Patients with PD may have postural and action tremors.

3. Rigidity. 89-99% patients with PD

- a) Increased resistance noted uniformly during range of passive joint movement.
- b) Can be enhanced by contralateral motor activity or mental task performance.

STEP TWO: Exclude Atypical Features

Once parkinsonism has been diagnosed:

Diagnosis of clinically established PD requires:

1. Absence of absolute exclusion criteria ([Table 1](#))
2. At least two supportive criteria ([Table 2](#)), and
3. No red flags ([Table 3](#))

Diagnosis of clinically probable PD requires:

1. Absence of absolute exclusion criteria ([Table 1](#))
2. Presence of red flags counterbalanced by supportive criteria, e.g.,
 - a) If 1 red flag is present, there must also be at least 1 supportive criterion.
 - b) If 2 red flags, at least 2 supportive criteria are needed.
 - c) No more than 2 red flags are allowed for this category.

Table 1. Absolute exclusion criteria. The presence of any of these features rules out PD.

1.	Cerebellar signs	Unequivocal cerebellar abnormalities, such as cerebellar gait, limb ataxia, or cerebellar oculomotor abnormalities (e.g., sustained gaze evoked nystagmus, macro square wave jerks, hypermetric saccades)
2.	Downward supranuclear palsy	or selective slowing of downward vertical saccades
3.	Frontotemporal dementia	Diagnosis of probable behavioural variant frontotemporal dementia or primary progressive aphasia, defined according to consensus criteria within the first 5 years of disease
4.	Purely lower limb parkinsonism	Parkinsonian features restricted to lower limbs for > 3 years
5.	Drug-induced	Treatment with dopamine receptor blocker or dopamine-depleting agent in a dose and time-course consistent with drug-induced parkinsonism
6.	No response to Levodopa	Absence of observable response to high-dose levodopa despite at least moderate severity of disease
7.	Cortical sensory	Unequivocal cortical sensory loss (i.e., graphesthesia, stereognosis with intact primary sensory modalities), clear limb ideomotor apraxia, or progressive aphasia
8.	Normal functional neuroimaging	of the presynaptic dopaminergic system
9.	Alternative diagnosis	Documentation of an alternative condition known to produce parkinsonism and plausibly connected to patient's symptoms, or the expert evaluating physician, based on the full diagnostic assessment feels that an alternative syndrome is more likely than PD

Table 2. ≥ 2 PD Supportive Features.

Good response to dopaminergic therapy	• Return to normal or near-normal level of function, or • During titration phase: Marked improvement with dose ↑ or marked worsening with dose ↓ [$> 30\%$ in Unified Parkinson's Disease Rating Scale (UPDRS) III, or subjectively with a clear history of marked changes], or • Unequivocal and marked on / off fluctuations, which must have at some point included predictable end-of-dose wearing off NB: Multiple System Atrophy – Parkinsonian type (MSA-P) may have partial Levodopa (L-dopa) response
Levodopa-induced dyskinesia	NB: MSA-P may have partial L-dopa response
Rest tremor	of a limb, documented on clinical examination (in the past, or on current examination)
Olfactory loss OR Cardiac MIBG ↓ uptake	in the anosmic or clearly hyposmic range, adjusted for age and sex Cardiac Metaiodobenzylguanidine (MIBG) scan clearly documenting cardiac sympathetic denervation

Table 3. ≤ 2 Red Flags

Within 3 years onset:		
1.	Recurrent falls	> 1 year, because of impaired balance
Within 5 years onset:		
2.	Rapid gait worsening	Requiring regular use of wheelchair
3.	Bulbar	Severe dysphonia or dysarthria (speech unintelligible most of the time) or severe dysphagia (requiring soft food, nasogastric tube or gastrostomy feeding)
4.	Autonomic	Severe autonomic failure. This can include: a) Orthostatic Hypotension - ↓ blood pressure within 3 minutes of standing by ≥ 30 mmHg systolic or 15 mmHg diastolic, in the absence of dehydration, medication, or other diseases that could plausibly explain autonomic dysfunction, or b) Severe urinary retention or urinary incontinence in < 5 years (excluding long-standing or small amount stress incontinence in women) that is not simply functional incontinence. In men, urinary retention must not be attributable to prostate disease and must be associated with erectile dysfunction
5.	No progression	A complete absence of progression of motor symptoms or signs unless stability is related to treatment
6.	No non-motor symptoms	Absence of any of the common NMS despite 5 years disease duration. These include sleep dysfunction [sleep-maintenance insomnia, excessive daytime somnolence, symptoms of Rapid Eye Movement – Behaviour Disorder (RBD)], autonomic dysfunction (constipation, daytime urinary urgency, symptomatic orthostasis), hyposmia, or psychiatric dysfunction (depression, anxiety or hallucinations)
Within 10 years onset:		
7.	Dystonia / contractures	Disproportionate anterocollis (dystonic) or hand / feet contractures
At any time:		
8.	Inspiratory signs	Inspiratory respiratory dysfunction: either diurnal or nocturnal inspiratory stridor or frequent inspiratory sighs
9.	Pyramidal signs	Otherwise-unexplained pyramidal tract signs, defined as pyramidal weakness or clear pathologic hyperreflexia (excluding mild reflex asymmetry and isolated extensor plantar response)
10.	Symmetric parkinsonism	The patient or caregiver reports bilateral symptom onset with no side predominance and no side predominance is observed on objective examination

Initial Assessment

Initial assessment involves diagnosis review, evaluation of risk factors and presence of any other co-morbid conditions (such as diabetes mellitus, hypertension and dyslipidaemia) or complications, basic health education and evaluation of learning needs and barriers.

Course of Parkinson's Disease

PD results in significant disability 10 to 15 years after its onset (but rate of disease progression may vary among patients).

Falls are common - result of postural instability, postural hypotension, dyskinesias, confusion, dementia, suboptimal nutrition, and sleep disorders.

Mean survival before days of Levodopa was 9 years, with a mortality ratio of 3.0 compared to the general population.

Bronchopneumonia and urinary tract infection were the common causes of death.

Levodopa provides effective treatment for ameliorating the motor symptoms in early PD and reduces the mortality ratio to 1.5 compared to the general population.

Table 4. Median disease duration to reach the various Hoehn and Yahr stages

Hoehn and Yahr stage	Clinical severity	Median disease duration (years) for untreated patients	Median disease duration (years) for levodopa-treated patients
I	Unilateral parkinsonism	3	Not available
II	Bilateral parkinsonism	6	9
III	Mild to moderate disability with postural impairment	7	12
IV	Severe disabling disease, able to walk unassisted but markedly incapacitated	9	12
V	Confined to bed or wheelchair unless aided	14	18

MANAGEMENT

PD is a chronic disease covered under the Chronic Disease Management Programme (CDMP) / Community Health Assist Scheme (CHAS).

Goals of Treatment

1. To control symptoms and to improve function and quality of life.
2. Balance between improving symptom but potentially inducing a drug side effect.
3. Seldom possible to abolish symptoms.

Pharmacotherapeutic Management

Types of pharmacologic treatment:

1. Correcting dopamine deficiency as a result of loss of dopaminergic neurons.
2. Increase availability of dopamine in the brain – Levodopa, the precursor of dopamine which is readily converted to dopamine by dopa decarboxylase.
3. Enhance effects of dopamine with dopamine agonist, or
4. Reduce Levodopa / dopamine breakdown with enzyme inhibitors.

Pharmacotherapeutic Management: Motor Symptoms

Drug	Common Dose	Common ADR	Remarks/ Contraindications/ Precautions
Levodopa Preparations Madopar® (Levodopa/Benserazide): 125 mg Capsules (S1) \$ 125 mg HBS Capsules (S2) \$ 250 mg Tablets (S1) \$ Levodopa/Carbidopa preparations: (Not available in NUP) Credanil® 100/25mg or 250/25mg Tablets Sinemet® CR 200/50mg Tablets	Start L-dopa at 62.5 mg TDS → 125 mg TDS → 187.5 mg TDS → 250 mg TDS (titrate at 1 - 2 weekly intervals with regular reviews to assess response and functional status). May titrate at an even slower rate if the patient is unable to tolerate standard titration.	Nausea, vomiting (may use domperidone but with caution according to MOH guidelines), postural hypotension, cognitive disturbance. Motor complications i.e. dyskinesia, motor fluctuations (sudden, unpredictable fluctuations between ON and OFF state). May cause sweat, urine and saliva to appear orange in colour.	- Main treatment. Improves bradykinesia and rigidity, but not so much tremor. - Usually administered with a peripheral dopa decarboxylase inhibitor, Carbidopa (Sinemet [not available in NUP]) or Benserazide (Madopar), in order to reduce the peripheral metabolism of levodopa. This enables more of it to cross the blood-brain barrier to reach the brain. - Slow release L-dopa (e.g., Madopar® HBS) is associated with an unpredictable ON period. However, it can be useful as the last dose before bedtime for patients with nocturnal OFF. Administration: Best to be given on empty stomach i.e., ½ hr before food or 1 hr after food. May be taken with low-protein food (like biscuits, fruit juice) if unable to tolerate GI side effects. Avoid taking with or after protein rich food as it may affect absorption. Madopar HBS can be taken with or without food. Drug Interactions: Space at least 2 hours apart from iron supplements. Avoid metoclopramide and olanzapine. Caution using first-generation antipsychotics (e.g., chlorpromazine, trifluoperazine, haloperidol) as it may worsen motor symptoms of PD.
Dopamine Agonists (DAs) (Not available in NUP) Non ergot-derived: Ropinirole Requip® 0.25mg, 1mg, 2mg Tablet Requip® PR 2mg, 4mg Tablet Rotigotine Patch Pramipexole Piribedil	Ropinirole: Initial 0.25mg TDS, usual 12-16mg daily in 3 divided doses, up to max 24mg/day in 3 divided doses	Nausea, drowsiness, hallucinations, insomnia, constipation, peripheral oedema. Rasagiline has significantly less of these side-effects.	- May be considered as first line medication for younger PD patients (age < 65 years) and should be titrated to a clinically efficacious dose. The use of long-acting DAs as monotherapy may delay the need for L-dopa and hence the onset of motor complications. However, do not hesitate to initiate L-dopa once subjectively needed as the main aim of treatment is patients' quality of life while they are still relatively young. - Use as add-on when motor complications set in. Considered if patient has no co-morbidity, no cognitive impairment, and has a considerable life expectancy. - Avoid the use of DAs in young males with a propensity to, or history of addiction disorders. Every patient who is initiated on a DA must be counselled on possible development of impulse control disorders such as compulsive sexual activity, eating, pathological gambling & shopping.
Monoamine Oxidase-B (MAO-B) Inhibitors Selegiline 5mg Tablet (NS) \$-\$\$ Rasagiline 1mg Tablet (Not available in NUP)	Selegiline: 2.5 - 5 mg OD or BD Rasagiline: 1mg OM	Headache, dry mouth, neuropsychiatric disturbance, nausea, dizziness, insomnia. Rasagiline has significantly less of these side-effects.	- May be considered as first line medication for younger PD patients with mild disease - Use as add-on when motor complications set in. Administration: For Selegiline BD dosing, take with breakfast and lunch to avoid insomnia.
Catechol-O-methyl transferase (COMT) inhibitors (Not available in NUP) Entacapone (Comtan® 200mg Tablet)	Entacapone: 200 mg with each dose of Levodopa, up to a maximum of 8 times daily (Max 1600mg/day)	Enhances L-dopa side effects. Orange urine discolouration.	Used to lengthen the ON period before the next dose of L-dopa is due. The additional ON time gained from each dose of Entacapone is around 15-45 min.

Drug	Common Dose	Common ADR	Remarks/ Contraindications/ Precautions
Stalevo® (Levodopa/Carbidopa/Entacapone 100/25/200mg Tablets)			
Anticholinergics Benzhexol 2mg Tablets (S1) \$ Bentropine 2mg Tablet (Not available in NUP)	Benzhexol: 2 mg BD-TDS Bentropine: 1-2 mg BD	Dry mouth, confusion, blurred vision, constipation.	• May be used for tremor-dominant PD in younger patients. • Anticholinergics do not significantly improve the other features of parkinsonism. • NOT to be used in elderly because of risk of cognitive decline and hallucinations.
Amantadine (Symmetrel® 100mg Tablet) (Not available in NUP)	Initial 100mg OM, maintenance 100mg OD- BD, max 400mg/day in divided doses	Livedo reticularis, peripheral oedema, confusion, hallucinations.	• May be used as add-on for treatment of Levodopa-induced dyskinesia. • Caution in patients with renal impairment.

^ Cost \$ <\$10 per month \$\$\$\$ \$30-<\$60 per month
 \$\$ \$10-<\$20 per month \$\$\$\$\$ \$60-<\$90 per month
 \$\$\$ \$20-<\$30 per month \$\$\$\$\$\$ \$90 or more per month

Amount payable depends on patient subsidy level and drug subsidy class (Standard Drug List S1 and S2, Non-Standard Drug NS)

Pharmacotherapeutic Management: Non-motor Symptoms

1. Neuropsychiatric Symptoms

- a) Depression – Pramipexole has anti-depressant effects. SSRIs may be considered to treat depression in Parkinson disease without dementia. (Selegiline should be stopped when SSRI started in view of risk of serotonergic crisis)
- b) Dementia - Donepezil or rivastigmine may be considered for PD patients with dementia.
- c) Psychosis - PD patients with severe psychosis should be referred to a psychiatrist and clozapine may be required. Clozapine has significant side effects and needs a strict regime of investigations and monitoring for leukopaenia. For milder cases, quetiapine is commonly used, but not olanzapine.

2. Autonomic Dysfunction

- a) Orthostatic hypotension – Increase fluid and salt intake. Midodrine and fludrocortisone may be used. (Side effects of fludrocortisone: hypertension, hypokalemia, ankle oedema). Midodrine is short-acting and may be a better choice in patients with supine hypertension as the timing of the last dose can be adjusted according to the patient's bedtime
- b) Gastrointestinal side effects
 - Constipation and reduced gastric motility are common in PD – regular pre-and pro-biotics, lactulose or forlax is effective. Avoid prolonged use of senna, bisacodyl and fleet enema
 - Anorexia, nausea and vomiting are common side effects of dopamine agonist therapy. Domperidone may be considered with caution according to MOH guidelines.
- c) Erectile dysfunction
 - May be treated with sildenafil, although the patient has to be forewarned of side effects like headaches, transient visual effects and flushing, and dangerous side effects like cardiac arrest and hypotension. Priapism is also known to occur.

Surgical Management

Deep brain stimulation of subthalamic nucleus or globus pallidus:

- Involves the placement of stimulating electrodes into the relevant nucleus to depolarize it. The electrodes are then connected to a pulse generator implanted subcutaneously below the clavicle, much like a cardiac pacemaker.
- Currently performed in Singapore @ National Neuroscience Institute.

Ancillary Management

Rehabilitation services comprising physical, occupational and speech therapy can do much to help patients who have gait difficulties, dysphonia, dysarthria or dysphagia to improve their quality of life. Where necessary, doctors should refer patients to dieticians for advance dietary advices, MSWs or psychologists for counselling, and physiotherapists for balance and strength training. As occupational and speech therapy services are not available in NUP, consider referring to Neurology and/or Rehabilitation Medicine for further assessment if PD patients develop any issues with daily functioning, oropharyngeal dysfunction, swallowing impairment or speech impairment.

Ancillary Management

Problem	Support Service	Management
General fatigue	Physical therapy	Exercise therapy
Gait difficulties / start hesitation / falls	Physical therapy	Gait training Strengthening exercises Visual auditory, and tactile cues Exercises to improve posture
Reduced pulmonary function	Physical therapy	Pulmonary and general exercises Exercises to improve posture
Difficulty with activities of daily living (work, leisure and self-care activities)	Occupational therapy	1. Group occupational therapy 2. Occupational aids 3. Review of home / work environment
Dysphonia	Speech therapy	1. Lee Silverman Voice Treatment (LSVT) 2. Pitch Limited Voice Treatment (PLVT)
Stuttering	Speech therapy	1. Prosody exercises 2. Chorus speech 3. Smooth speech 4. Delayed auditory feedback
Tachyphemia and problems with prosody (intonation, rhythm, and lexical stress in speech)	Speech therapy	1. Prosody exercises 2. Smooth speech
Dysarthria	Speech therapy	1. Lee Silverman Voice Treatment (LSVT)
Dysphagia	Speech therapy	Investigation and intervention (thickeners) May require feeding via nasogastric tube or percutaneous endoscopic gastrostomy

Clinical Quality

Good Practices

1. For every new drug prescribed for treatment of PD, check and document response to the therapy and occurrence of side effects.
2. Antiparkinsonian medication should not be withdrawn abruptly or allowed to fail suddenly owing to poor absorption (e.g., gastroenteritis, abdominal surgery), to avoid the potential for neuroleptic malignant-like syndrome (Parkinsonism hyperpyrexia syndrome).
3. Ask about activity level and recent falls. Promote physical activity.
4. Look out for neuropsychiatric symptoms – depression, psychosis, dementia.
5. Look out for autonomic dysfunction – orthostatic dizziness / hypotension; constipation, anorexia, nausea, vomiting; erectile dysfunction.
6. For patients on clozapine for psychosis, monitor for leukopenia.
7. Look out for caregiver stress.

Role of Health Team Members

1. Family Physician

- Able to provide all aspects of primary medical care from screening, diagnosis and management of PD, including health promotion and prevention / treatment of complications.
- Collaborates with other health care providers in the health team to provide holistic and patient-centred care.

2. Care Coordinator

- Introduce OneNUHS and HealthHub Apps to patient.
- Introduce and enrol patient onto Teleconsult.
- Perform PHQ2, frailty and functional screenings, abbreviated mental test (AMT) and comprehensive geriatric assessment (CGA) when indicated.
- Address care gaps under Health Maintenance Topics, such as vaccinations (influenza / pneumococcal), cancer screenings (FIT kit, mammogram, PAP smear / HPV), DRP and DFS.

3. Advanced Practice Nurse

- Co-manage patients who are diagnosed with mild and stable PD (e.g., symptoms review, post-medication review) including pre-existing chronic conditions with physician and health team.
- Liaise with other healthcare personnel to close care gaps and provide holistic and patient-centred care.

4. Financial Counsellor

- Financial counselling for patients with issues on payment of medical bills.
- Advise on MediSave utilisation for chronic disease medical bills.
- Application on government endowment fund- MediFund and Medication Assistance Fund for subsidies on high-cost drugs.
- Application on Financial Assistance Scheme for eligible non-residents.

5. Dietitian

- Dietary management for patients with malnutrition or poor appetite / oral intake.
- Dietary management for patients with swallowing difficulties/ enteral feeding.

6. Medical Social Worker

- Conduct biopsychosocial assessment.
- Provide intervention for social assistance including application of schemes and services, and referral for community resource and caregiver support.
- Consider alternative financial support (not related to medical bills) when necessary.
- Referrals to PD related care in the community when needed.

7. Psychologist

- Emotional and psychological support to facilitate adjustment to newly diagnosed PD, or to manage change in the disease.
- Psychological assessment and intervention to manage mood challenges (depression, anxiety etc.) and behavioural issues associated with PD.
- Support patient and caregiver in developing strategies to improve daily functioning and quality of life.
- Psychological support for carers experiencing caregiver stress.

8. Physiotherapist

- Patient education regarding physical activity and reducing sedentary periods.
- Patient education regarding exercising within the ON time of dopaminergic therapy and the use of visual auditory and tactile cueing to reduce freezing.
- Exercise prescription on resistance, stretching, balance, co-ordination, and aerobic exercises to improve Parkinson's/motor symptoms.
- Exercise prescription on strengthening and stretching to improve upright postures.

9. Podiatrist

- To manage painful pathological callus and nails.
- Footwear and foot care education.

REFERRAL TO SPECIALIST

When to Refer to Specialist

1. New diagnosis of PD – refer to neurologist if unsure of diagnosis or starting medication.
2. Young-onset PD.
3. Atypical Parkinsonian disorders.
4. Patients with family history of PD.
5. Patients who do not respond to levodopa or dopamine agonists.
6. Patients with cognitive impairment or neuropsychiatric dysfunction.
7. Motor complication (i.e., motor fluctuations and dyskinesias) not responding to medication adjustments.

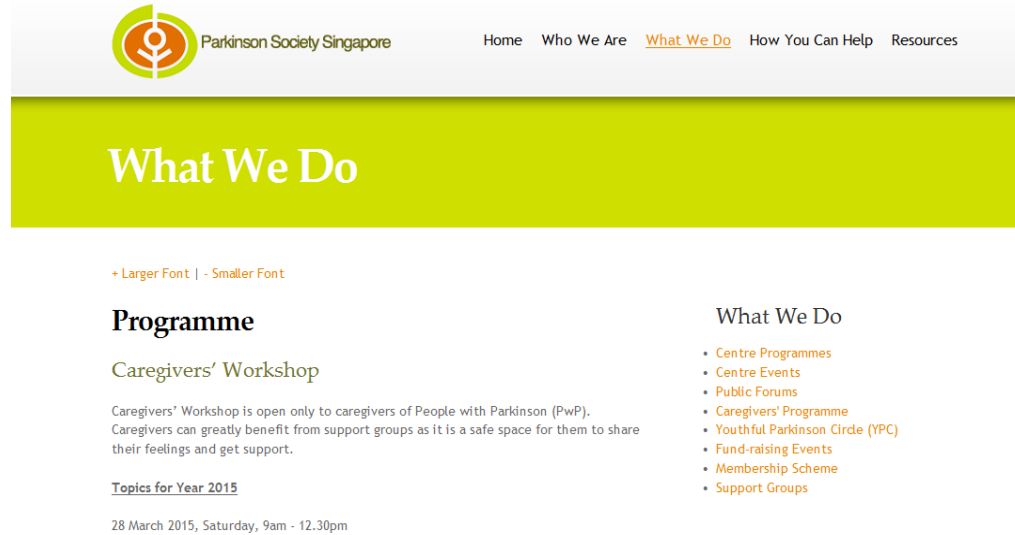
SPECIAL SITUATIONS

Fitness to Drive

1. If a medical concern is raised regarding the patient's fitness to drive, he should be advised to stop driving and referred to the neurologist for further assessment and a driving test.
2. The patient is deemed unfit to drive if the following are of sufficient degree to interfere with safe driving:
 - a) Significant weakness
 - b) Lack of coordination
 - c) Involuntary movements
 - d) Visual impairment

Useful Links for Patients and Caregivers

1. The Parkinson's Disease Society (Singapore): www.parkinsonsingapore.com
2. WE MOVE TM (Worldwide Education and Awareness Movement Disorders): www.wemove.org
3. U.S. National Parkinson Foundation: www.parkinson.org



The screenshot shows the Parkinson Society Singapore website. The header includes the logo and navigation links: Home, Who We Are, What We Do (highlighted), How You Can Help, and Resources. Below the header is a large green banner with the text 'What We Do'. Underneath the banner, there are two columns. The left column is titled 'Programme' and features a section for 'Caregivers' Workshop' with a description and a link to 'Topics for Year 2015'. The right column is titled 'What We Do' and lists several programs: Centre Programmes, Centre Events, Public Forums, Caregivers' Programme, Youthful Parkinson Circle (YPC), Fund-raising Events, Membership Scheme, and Support Groups.

Programme

Caregivers' Workshop

Caregivers' Workshop is open only to caregivers of People with Parkinson (PwP). Caregivers can greatly benefit from support groups as it is a safe space for them to share their feelings and get support.

[Topics for Year 2015](#)

28 March 2015, Saturday, 9am - 12.30pm

What We Do

- Centre Programmes
- Centre Events
- Public Forums
- Caregivers' Programme
- Youthful Parkinson Circle (YPC)
- Fund-raising Events
- Membership Scheme
- Support Groups

RECOMMENDED CARE COMPONENTS FOR PARKINSON'S DISEASE

Recommended Care Components	Minimum Frequency*	Remarks
Review of Diagnosis	Annually	The diagnosis would be reviewed regularly and reassessed if there are atypical features (e.g., falls at presentation and early in the disease course, poor response to levodopa, symmetry at onset, rapid progression to Hoehn & Yahr stage 3 in 3 years, lack of tremor or dysautonomia)
Review of Treatment	Annually	Review and discussion regarding medical and surgical treatment options, as well as need for rehabilitative therapies (physiotherapy, occupational therapy and speech therapy)
Review of Complications	Annually	Assessment for cognitive impairment, psychiatric disorders (e.g., depression, psychosis), autonomic dysfunction (e.g. constipation, incontinence, orthostatic hypotension), falls, sleep disorders, and medication-related side effects
Influenza Vaccination [#]	Annually or per season	As recommended under the National Adult Immunisation Schedule (NAIS)

** More frequently if clinically indicated*

This is a non-reportable clinical indicator

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